Architectural Skills

CSSE6400

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Quote

Architecture is the stuff you can't Google.

– Mark Richards [Richards and Ford, 2020]

Quote

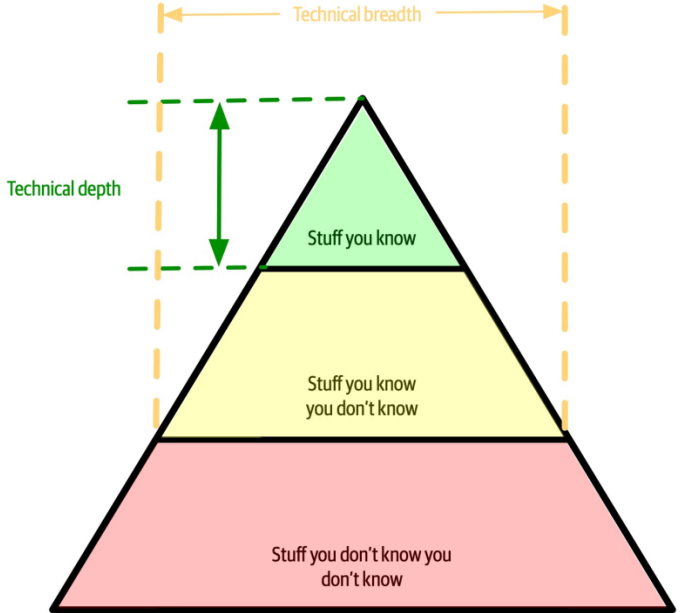
There are no right or wrong answers in architecture—only trade-offs.

– Neal Ford [Richards and Ford, 2020]

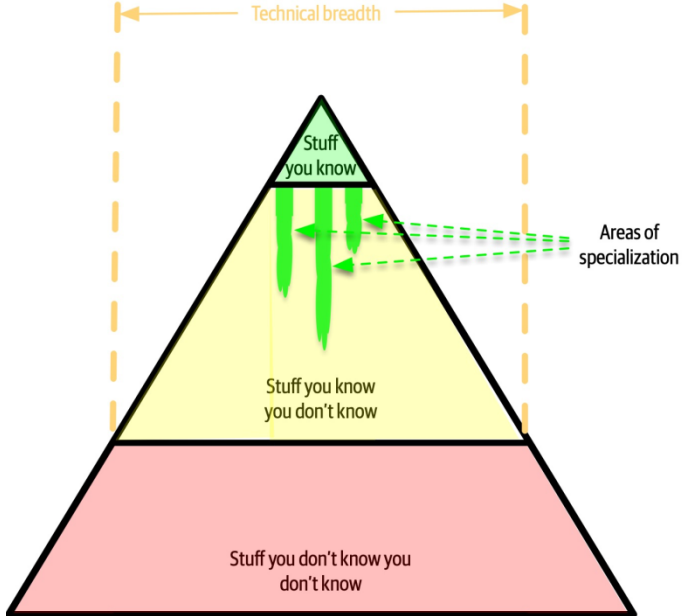
Architectural Design

Architects use knowledge and experience to analyse trade-offs to design architectures appropriate to the system context.

Developers – Technical Depth [Richards and Ford, 2020]



Architects – Technical Breadth [Richards and Ford, 2020]



- Architects need greater technical breadth than depth
- Breadth allows better consideration of trade-offs
- Avoid trying to become an expert across many areas
  - You'll fail
- Don't stop learning – increase your breadth
  - Don't let your knowledge become stale

Definition 0. Conway's Law

Organisations design systems whose structure is inevitably a copy of the organisation's communication structure[Conway, 1968] [MacCormack et al., 2012].

- First citation is original article.
- Second is one of several about MIT and Harvard research into the phenomenon, calling it the *mirroring hypothesis*.
  - Compared open source to commercial packages (e.g. Linux to Solaris).
- Elaborate on this point and Coplien's research into organisational sociology.
- Reasons
  - Governance structures constrain communication paths.
  - This constrains space in which to search for solutions, constraining problem solving approaches.

Conway's Law Consequences

- Business Process Management
- Microservices to reflect organisation structure
- Teams formed around services

- BPM: Redesign organisation structure to reflect system you want
- Microservices: Design system to reflect your organisation
- Elaborate on benefits of both approaches
- Comment on benefits of small focused teams

Conway's Law Consequences

Team insularity – more loyal to team than organisation

- Amazon order tracking example from 2022 week 11
  - Negotiation difficulties with other teams
- Need to ensure inter-team cooperation
- Possibly move people between teams
- Microservices can encourage insularity
- Many teams can also encourage insularity
- Intra-team communication becomes more difficult with large teams

Conway's Law Issues

- Cross-cutting concerns
  - e.g. Security
- Organisation structure should align with market structure
- Physical location of teams

- Cross-cutting concerns span services & consequently teams
- Can't have a "security" service
  - Needs to be part of *every* service
- Teams solely based around Conway's law and services may not deliver some cross-cutting concerns
- Cooperation, documentation & audits may be necessary
- Market structure may complement team structure to place teams closer to their end users
- Global development & outsourcing mean different teams are likely to be in different locations
- Requires additional overhead & documentation for cooperation between teams

Definition 0. Peopleware[Neumann, 1977]

People involved in development of systems.
– Peter G. Neumann

Stakeholders [Coplien and Bjørnvig, 2010]

- End Users
- Organisation
- Customers / Sponsors
- Domain Experts
- Developers

- Organisation for whom system is being developed
- Customers / Sponsors are those who are responsible for the end user's work process (e.g. managers)
- Domain Experts understand details of problem and/or solution domain

- Written
  - For those who are not there
- Oral
  - Immediate & interactive

- Both apply to all types of stakeholders
- Time Dimension: Written is available in the future
- Extent of either depends on SE process
- Extent also depends on engagement of each type of stakeholder
- Technical Documentation guest lecture in week 11 only touched on docs written for developers
  - Points about clarity are still relevant

Circumventing Conway's Law [Woods, 2017]

- Cloud Platforms
- Microservices
- APIs
- *Culture*

- Culture is most important
- Easily deployed & accessible services with good APIs opens possibility of communication outside of organisation silos.

Evidenced-Based Software Engineering

Don't follow fads, seek evidence for good practice.

Elaborate on finding reliable sources of information and confirming facts yourself.

Let's hear from an expert

Software Engineering's Greatest Hits

what we actually know about software development and why we believe it's true

Greg Wilson

http://third-bit.com/talks/greatest-hits/

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https://youtu.be/HrVtA-ue-x0

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